CSE STUDY HUB

Emerging Technology

UNIT 5 · Sustainable & Future Technologies

What this unit is about

This unit builds understanding of **Sustainable & Future Technologies**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Renewable energy: solar, wind, hydro and storage

2. Smart grids, demand management and distributed energy

3. Sustainable materials, recycling and circular economy

4. Green computing: energy-efficient hardware and data centres

5. Environmental impact assessment and lifecycle thinking

6. Emerging trends: batteries, hydrogen, carbon capture and smart mobility

7. Trade-offs: cost, scale, resource use and social acceptance

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 5 NOTES · PART 1

Renewable energy: solar, wind, hydro and storage

Renewable energy: solar, wind, hydro and storage is a syllabus concept in Emerging Technology. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Smart grids, demand management and distributed energy

Smart grids, demand management and distributed energy is a syllabus concept in Emerging Technology. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Sustainable materials, recycling and circular economy

Sustainable technology aims to meet present needs while reducing long-term environmental and social costs. Assess energy/material use across the lifecycle, not only during operation.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Sustainable & Future Technologies, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 5 NOTES · PART 2

Green computing: energy-efficient hardware and data centres

Green computing: energy-efficient hardware and data centres is a syllabus concept in Emerging Technology. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Environmental impact assessment and lifecycle thinking

Present Environmental impact assessment and lifecycle thinking as an ordered workflow. State the preconditions, list the major steps, explain what each step produces, and mention how errors are detected or handled. Use a small worked example or flowchart in the exam.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Sustainable & Future Technologies, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 5 NOTES · PART 3

Emerging trends: batteries, hydrogen, carbon capture and smart mobility

Emerging trends: batteries, hydrogen, carbon capture and smart mobility is a syllabus concept in Emerging Technology. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Trade-offs: cost, scale, resource use and social acceptance

Compare the alternatives in Trade-offs: cost, scale, resource use and social acceptance using the same criteria: purpose, working method, performance, cost/complexity, security or reliability, and suitable use case. End with a short conclusion stating when each option is preferable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Sustainable & Future Technologies, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Renewable energy: solar, wind, hydro and storage** — know the definition, core working and one example.

• **Smart grids, demand management and distributed energy** — know the definition, core working and one example.

• **Sustainable materials, recycling and circular economy** — know the definition, core working and one example.

• **Green computing: energy-efficient hardware and data centres** — know the definition, core working and one example.

• **Environmental impact assessment and lifecycle thinking** — know the definition, core working and one example.

• **Emerging trends: batteries, hydrogen, carbon capture and smart mobility** — know the definition, core working and one example.

• **Trade-offs: cost, scale, resource use and social acceptance** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Sustainable & Future Technologies. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Renewable energy: solar, wind, hydro and storage and explain its importance.

Q2. Explain the working/steps involved in Smart grids, demand management and distributed energy.

Q3. Differentiate between two related concepts from Sustainable & Future Technologies.

Q4. Give one practical application of Trade-offs: cost, scale, resource use and social acceptance and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Sustainable & Future Technologies.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.